

# Fuzhao XUE

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## INDUSTRY EXPERIENCE

### **Google DeepMind | London    *Senior Research Scientist***

Dec. 2024-Present

*In charge of efficient pre-training for Gemini Project.*

- Working on model architecture, distillation, scaling science for Gemini efficient large-scale pretraining.
- Core-contributor of the Gemini 2.5, Gemini 3.0, Gemini 3.5 models.

### **NVIDIA | Work Remotely from Singapore    *Research Intern***

Jan. 2023-Sep 2024

*In charge of multi-model foundation model pretraining and context scaling. Working with Linxi(Jim) Fan and Yuke Zhu.*

- Trained video language model for general video captioning with more high-quality data and compute.
- Designed and implemented infrastructure to train and deploy long context video language model.

### **Google | Work Remotely from Singapore    *Student Researcher***

Jul. 2022-Nov 2022

*In charge of adaptive computation with adaptive model depth and elastic input sequence. Worked with Yi Tay and Mostafa Dehghani. Part of the work is accepted at ICML 2023.*

- Implemented universal transformer and PonderNet on vision transformer based on Scenic and made vision universal transformer open source.
- Proposed AdaTape to achieve adaptive computation with elastic input sequence, which is a novel strategy that enables dynamic computation in neural networks via adaptive tape token. (ICML 2023)

## EDUCATION

### **Doctor of Philosophy (Ph.D.)**

Oct. 2024

National University of Singapore

Singapore

- Majored in Computer Science
- Supervisor: Yang You
- Google Ph.D. Fellowship
- President's Graduate Fellowship
- AAAI 2023 Distinguished Paper Award (Co-author)
- Published papers at ICML, NeurIPS, ACL, AAAI, CVPR
- Internship at Google (Brain/DeepMind Team) and worked with Yi Tay and Mostafa Dehghani
- Internship at NVIDIA (GEAR Team) and worked with Linxi (Jim) Fan and Yuke Zhu
- Thesis: Towards Efficient Transformer Scaling

### **Master of Engineering (M.Eng.)**

Jul. 2021

Nanyang Technological University

Singapore

- Majored in Computer Science and Engineering
- Supervisor: Eng-Siong Chng and Aixin Sun
- Published paper at AAAI, ICML and ICASSP.
- Thesis: Refining latent multi-view graph for relation extraction
- GPA: 5.0/5.0

## RESEARCH INTEREST

- **Machine Learning:** Foundation Model Scaling, Conditional Computation, Adaptive Computation
- **Natural Language Processing:** Large Language Model Pre-training and Instruction Tuning
- **High Performance Computing:** Model Parallelism and Large-scale Deep Learning System for Long Sequence.

## HIGHLIGHT RESEARCH

- **OpenMoE:** The FIRST FULLY OPEN MoE-based Decoder-only LLM Trained over Chinchilla Scaling Law.
  - OpenMoE: An Early Effort on Open Mixture-of-Experts Language Models. ICML 2024
    - ✧ **Fuzhao Xue**, Zian Zheng, Yao Fu, Jinjie Ni, Zangwei Zheng, Wangchunshu Zhou and Yang You
- **Token Crisis:** The FIRST Project Studying Training LLM with Limited and Repeated Data
  - To Repeat or Not To Repeat: Insights from Scaling LLM under Token-Crisis. NeurIPS 2023
    - ✧ **Fuzhao Xue**, Yao Fu, Wangchunshu Zhou, Zangwei Zheng, Yang You
- **AdaTape:** Adaptive Computation Foundation Model Supporting Elastic Input Sequence and Dynamic Read&Write,
  - Adaptive Computation with Elastic Input Sequence. ICML 2023
    - ✧ **Fuzhao Xue**, Valerii Likhoshesterov, Anurag Arnab, Neil Houlsby, Mostafa Dehghani, Yang You

- **Sequence Parallelism and Ring Self-Attention:** The FIRST Work Trying to Solve Long Sequence Training with Distributed System Instead of Improving Attention Layer Efficiency.
  - Sequence Parallelism: Long Sequence Training from System Perspective, ACL 2023
    - ✧ Shenggui Li\*, **Fuzhao Xue\***, Yongbin Li, Yang You
- **MixEval:** The FIRST Work Mixing Ground-truth-based Benchmarks and achieving 0.96 correlation with Chatbot Arena.
  - MixEval: Deriving Wisdom of the Crowd from LLM Benchmark Mixtures
    - ✧ Jinjie Ni\*, **Fuzhao Xue\***, Xiang Yue\*, Yuntian Deng, Mahir Shah, Kabir Jain, Graham Neubig, Yang You
- **WideNet:** The SECOND Most Cited MoE-based Vision Transformer based on Google Scholar until Mar 2024
  - Go Wider Instead of Deeper, AAAI 2022
    - ✧ **Fuzhao Xue**, Ziji Shi, Futao Wei, Yong Liu, Yang You
- Full Publication List: <https://xuefuzhao.github.io/publications/>

## ACADEMIC RESEARCH EXPERIENCE

**National University of Singapore | Singapore** *Ph.D. Candidate* Jul. 2021-Present

*In charge of efficient and effective general training framework for both computer vision and natural language processing to scale transformer. Published at NeurIPS 2023, ICML 2023, ACL 2023, AAAI 2022.*

- Studied the relationship between transformer configuration and training objectives. (ICML 2023)
- Proposed sequence parallelism to train transformer with longer sequence from system perspective. (ACL 2023)
- Proposed to go wider instead of deeper, compressing along depth by parameter sharing and scaling along width by mixture-of-experts to construct a parameter-efficient framework. (AAAI 2022)

**Nanyang Technological University | Singapore** *M.Eng. Student* Jul. 2020-Jun. 2021

*In charge of design of improving dialogue-level relation extraction by adaptive graph pooling proposed. Published paper at AAAI 2021, ICASSP 2022.*

- Identified inductive words using Dynamic Time Wrapping Pooling in an unsupervised manner with high accuracy.
- Improved the state-of-the-art of dialogue-level relation extraction by 6% on DialogRE.

**National University of Singapore | Singapore** *Research Intern* Oct. 2019-Apr. 2020

*In charge of designing hybrid speech recognition framework with Pytorch and Kaldi. Published paper at ICML 2020.*

- Used Pytorch underlying operations to implement acoustic models such as DNN, LSTM, SRU, RRN and RTN.
- Modeled relational thinking via Deep Graph Process, reducing WER by 10% relatively on Chime2&5 and SWBD.

## SKILLS

- **Programming:** Python, C/C++, C#, Matlab
- **Frameworks:** PyTorch, JAX, TensorFlow, Keras, Theano, Scikit-learn, pandas
- **Advanced:** T5x, Flaxformer, Scenic, SeqIO, Deepspeed, Megatron, Huggingface

## SERVICES

- **Conference 2025:** ICLR Reviewer, AAAI Reviewer, CVPR Reviewer, ICML Reviewer
- **Conference 2024:** ICLR Reviewer, ICML Reviewer, COLM Reviewer
- **Conference 2023:** EMNLP Reviewer, NeurIPS Reviewer, ACL PC Member, CVPR Reviewer
- **Conference 2022:** EMNLP Reviewer, ICML Reviewer, SIGIR PC Member,
- **Journal:** TKDE Reviewer